MGU-UGP (HONOURS)

FOURTH SEMESTER MODEL EXAMINATION

(2024 ADMISSION ONWARDS)

MG4SECCND200– FOOD ADULTERATION

Duration: 2hrs Maximum Marks: 50

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

Each question is tagged with the appropriate Bloom's Revised Taxonomy level (Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating.) and the relevant course outcome (CO) number.

* Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S).

Part A

Multiple Choice Questions

Answer **All** Questions

Each question carries **1** mark

1. Which of the following is an example of an incidental adulterant? [K] [1]
   a) a) Metanil yellow added to turmeric b) Argemone oil mixed with mustard oil
   b) c) Pesticide residues present in fruits d) Chalk powder added to sugar
2. Which of the following is a non-permitted synthetic food colour? [K] [3]
   a) Tartrazine b) Sunset yellow
   c) Metanil yellow d) Erythrosine
3. The presence of lead chromate in turmeric powder is detected mainly to identify [K] [4]
   a) Artificial colouring agent b) Preservative
   c) Sweetener d) Antioxidant
4. Starch adulteration in milk can be detected using [K] [4]
   a) Benedict’s test b) Iodine test
   c) Fehling’s test d) Biuret test

5. Which of the following is a flavour enhancer? [K] [2]

a) Aspartame b) Lecithin

c) Monosodium glutamate d) BHA

6. Which additive is used as a raising agent in bakery products? [K] [2]

a) Sodium benzoate b) Baking powder

c) Citric acid d) Lecithin

[1 X 6 = 6]

Part B

Short Answer Type Questions

Answer **2** Questions

Each question carries **2** marks

7.Distinguish between intentional and incidental adulterants with one example each. [U] [1]

8. Briefly explain chemical food hazards and mention one control measure. [U] [1]

9.Distinguish between natural and synthetic food colours [U] [3]

10.What is HACCP? Mention its significance in food safety. [U] [1]

[2 X 2 = 4]

Part C

Short Essay Type Questions

Answer **4** Questions

Each question carries **5** marks

11. Explain the concept of food adulteration and classify the different types of adulterants with examples.[An] [1]

12. Describe the methods of detection of adulterants in milk. [A] [4]

13. Write short notes on PFA Act and its significance in food safety[U] [1]

14. Explain the functions of emulsifiers, stabilizers and glazing agents with examples[U] [2]

15. Write short notes on permitted and non-permitted food colours[U] [3]

16. Describe high fructose corn syrup (HFCS) and discuss its health effects. [An] [3]

[5 X 4 =20]

Part D

Essay Type Questions

Answer **2** Questions

Each question carries **10** marks

17. Explain the concept of food hazards. Describe the biological, chemical and physical hazards, their impact on human health, and the control measures adopted in food industries.[An] [1]
18. Discuss the common adulterants found in cereals, pulses, milk and spices. Explain the simple methods used for their detection.[An] [4]
19. Describe food colourants and classify them into natural and synthetic colours. Discuss the health hazards of synthetic food colours and their regulatory control.[E] [3]
20. Explain food sweeteners and classify them into natural and artificial sweeteners. Discuss their uses, relative sweetness and health implications.[E] [3]

[10 X 2 =20]